# THE STRONG COMPLETION TOWER: BAR-COBAR DUALITY FOR NON-QUADRATIC CHIRAL ALGEBRAS

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Abstract. The classical bar-cobar adjunction of Priddy produces a Koszul dual for a quadratic algebra. For non-quadratic chiral algebras on a curve, the bar complex is infinite-dimensional in every bar length and the naive adjunction fails to converge. We introduce the *strong completion tower*: a filtered augmented curved chiral $A_\infty$-algebra whose operations are filtration-nondecreasing on the augmentation ideal. Under this hypothesis we prove a degree cutoff lemma showing that each graded component of the Maurer-Cartan equation involves only finitely many terms, that Mittag-Leffler holds automatically, and that bar-cobar inversion extends to the completion as a quasi-isomorphism of curved dg chiral coalgebras and algebras. The result upgrades the previously conjectural MC4 completion closure to a theorem with sharp hypotheses and splits the completion problem into a positive regime ($\mathrm{MC4}^+$), unconditionally solved, and a weight-zero resonant regime ($\mathrm{MC4}^0$), tractable for finite resonance rank. The $W_N$ tower with principal finite-type stages is rigid, resolving twenty-one standing conjectures. Virasoro, affine Kac-Moody at non-critical level, principal $\mathcal{W}$-algebras, and lattice vertex algebras are strong completion towers.

## 1. Introduction

1.1. **The quadratic case and its limits.** Classical Koszul duality is a phenomenon of graded algebras. A quadratic algebra $A = T(V)/(R)$ determines a dual coalgebra $A^{!`} = T^c(sV)/(sR)$; the bar construction $B(A)$ mediates between them; and $A$ is Koszul when $H^*(B(A))$ concentrates in bar length one, so that $\Omega(A^{!`}) \simeq A$. The theory is due to Priddy [5], Beilinson-Ginzburg-Soergel, and in operadic form Loday-Vallette [2].

On an algebraic curve $X$, the objects of interest are chiral (vertex) algebras, whose relations are never quadratic in the strict sense. Heisenberg has a central term in the $J \cdot J$ OPE. Virasoro has a quartic pole. Principal $\mathcal{W}$-algebras $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ have OPE poles of order up to $2\,\mathrm{rk}(\mathfrak{g}) + 2$. Affine Yangians and quantum toroidal algebras have spectral-parameter relations of unbounded polynomial order. In each case the bar complex is a perfectly good

*Date*: April 13, 2026.

2020 *Mathematics Subject Classification.* 17B69, 18M70, 14H10, 81T40.

*Key words and phrases.* Chiral algebra, bar construction, Koszul duality, Maurer-Cartan, curved $A_\infty$-algebra, completion, Mittag-Leffler, $W$-algebra.

dg coalgebra, but the iterated OPE produces bar elements whose differential contains infinitely many summands. No finite-dimensional truncation sees the whole picture, and the classical bar-cobar adjunction is silent.

1.2. **The completion problem.** The remedy is completion. One introduces a decreasing filtration $\mathcal{A} = F^0\mathcal{A} \supset F^1\mathcal{A} \supset \cdots$ with finite-dimensional quotients $\mathcal{A}_{\leq N} := \mathcal{A}/F^{N+1}\mathcal{A}$ and writes $\widehat{\overline{B}}(\mathcal{A}) := \varprojlim_N \overline{B}(\mathcal{A}_{\leq N})$. At each finite stage $N$ the strict bar-cobar adjunction produces a universal twisting morphism $\tau_N \colon \overline{B}(\mathcal{A}_{\leq N}) \to \mathcal{A}_{\leq N}$ which is a Maurer-Cartan element of the convolution dg Lie algebra $\mathrm{Conv}(\overline{B}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N})$. The completion problem, labelled MC4 in [3], asks: do the finite-stage MC elements $\tau_N$ glue to a single MC element $\widehat{\tau}$ in the completed convolution algebra, and is the twisted tensor product acyclic in the limit?

Two obstructions appear. First, the MC equation $\partial(\widehat{\tau}) + \widehat{\tau} \star \widehat{\tau} = 0$ is a sum over all bar degrees, and a priori each component is infinite. Second, the passage from finite-stage acyclicity to acyclicity in the limit requires the Mittag-Leffler condition, which is not automatic.

1.3. **The main result.** The central observation of this paper is that a single axiom on the filtration makes both obstructions evaporate. The axiom is that the $A_\infty$ operations of $\mathcal{A}$ respect the additive weight:

$$\mu_r(F^{i_1}\mathcal{A}, \ldots, F^{i_r}\mathcal{A}) \subset F^{i_1+\cdots+i_r}\mathcal{A}. \tag{1.1}$$

A filtered curved chiral $A_\infty$-algebra for which (1.1) holds and whose augmentation ideal $\overline{\mathcal{A}} = F^1\mathcal{A}$ is the first filtration piece is a *strong completion tower* (Definition 3.1).

**Theorem 1.1** (Main theorem, informal statement)**.** *Let $\mathcal{A}$ be a strong completion tower whose finite-stage quotients lie in the proved bar-cobar regime. Then the completed bar construction $\widehat{\overline{B}}(\mathcal{A})$ is a separated complete pronilpotent curved dg chiral coalgebra, the completed convolution algebra carries a canonical Maurer-Cartan element $\widehat{\tau} = \varprojlim_N \tau_N$, Mittag-Leffler holds automatically, and the counit*

$$\widehat{\epsilon} \colon \widehat{\Omega}\big(\widehat{\overline{B}}(\mathcal{A})\big) \xrightarrow{\sim} \mathcal{A}$$

*is a quasi-isomorphism. Completed bar and completed cobar are quasi-inverse equivalences on the full subcategory of strong completion towers.*

The precise statement is Theorem 6.1 below. The proof assembles three ingredients: (i) the degree cutoff lemma (§4), which reduces each component of the MC equation to a finite sum; (ii) the automatic Mittag-Leffler property (§5), a direct consequence of the strong filtration axiom; and (iii) the reduction principle of Lemma 5.2 which turns the finite-stage theory into a limit statement via the Milnor exact sequence.

1.4. **Positive versus resonant towers.** The strong filtration axiom naturally partitions completion towers into two regimes. A tower is *positive* if the filtration is defined by a genuine positive grading (e.g. the $L_0$-eigenvalue filtration on a graded vertex algebra). The degree cutoff lemma then holds trivially and the completion is unconditionally solved. We call this the *MC4*$^+$ *problem*: it is now closed.

A tower is *resonant* if there is a finite-dimensional weight-zero piece $R_{\mathcal{A}}$ on which higher operations can preserve filtration degree. The *resonance rank*

$$\rho(\mathcal{A}) := \dim H^*(R_{\mathcal{A}}, m_1^{R_{\mathcal{A}}})$$

classifies the completion difficulty. Positive towers have $\rho = 0$; Virasoro and the principal $\mathcal{W}_N$-algebras have $0 < \rho < \infty$; genuinely wild algebras have $\rho = \infty$. The MC4$^0$ problem concerns the middle regime. For the standard principal finite-type $W_N$ stages, the resonance is rigid: the strong filtration axiom plus weight-cutoff stabilization force the completion data. This $W_N$ rigidity resolves twenty-one standing conjectures in the completion frontier, catalogued in §8.

1.5. **Relation to prior work and organization.** Positselski [4] developed curved bar constructions for associative algebras with curvature. Loday-Vallette [2] supply the operadic bar-cobar machinery on which the chiral lift is built. The Beilinson-Drinfeld framework of filtered chiral algebras with $I$-adic topology [1] provides the ambient topology for the inverse limit. The present theorem is the chiral and $A_\infty$ lift: the novelty is the strong filtration axiom, which identifies the precise structural hypothesis under which all finite-stage data assemble without further input.

Section 2 fixes notation. Section 3 defines the strong completion tower. Section 4 proves the degree cutoff lemma. Section 5 proves Mittag-Leffler. Section 6 assembles the main theorem. Section 7 discusses the MC4$^+$ versus MC4$^0$ dichotomy and the $W_N$ rigidity. Section 8 works out the standard families.

## 2. The bar construction on the augmentation ideal

Throughout, $X$ is a smooth complex algebraic curve and $\mathcal{A}$ is an augmented curved chiral $A_\infty$-algebra on $X$ in the sense of the monograph [3]. Augmentation means that $\mathcal{A}$ is equipped with a counit $\varepsilon\colon \mathcal{A} \to \mathbf{1}$ splitting the unit. The augmentation ideal is

$$\overline{\mathcal{A}} := \ker(\varepsilon).$$

*Remark* 2.1 (Augmentation ideal). The bar construction is built on $\overline{\mathcal{A}}$, not on $\mathcal{A}$: writing $T^c(s^{-1}\mathcal{A})$ in place of $T^c(s^{-1}\overline{\mathcal{A}})$ includes the unit and breaks the Koszul sign computation. We therefore set

$$\overline{B}^{\mathrm{ch}}(\mathcal{A}) := T^c\big(s^{-1}\overline{\mathcal{A}}\big),$$

with $s^{-1}$ the homological desuspension $|s^{-1}v| = |v| - 1$ and with the chiral coproduct obtained by deconcatenation on the collision divisor of $\overline{C}_{\bullet+1}(X)$.

The curved chiral $A_\infty$ operations $\{\mu_r\}_{r\geq 0}$ assemble into a codifferential $d_{\overline{B}}$ on $\overline{B}^{\mathrm{ch}}(\mathcal{A})$ with curvature coming from $\mu_0$, so $\overline{B}^{\mathrm{ch}}(\mathcal{A})$ is a curved dg coalgebra. The convolution dg Lie algebra $\mathrm{Conv}(\mathcal{C}, \mathcal{A}) := \mathrm{Hom}(\mathcal{C}, \mathcal{A})$ has bracket from the coproduct on $\mathcal{C}$ and the product on $\mathcal{A}$; a Maurer-Cartan element is a degree-1 map $\tau\colon \mathcal{C} \to \mathcal{A}$ with $\partial(\tau) + \tau \star \tau = 0$.

**Theorem 2.2** (Finite-type bar-cobar duality, cited)**.** *Let $\mathcal{A}$ be a finite-type augmented curved chiral $A_\infty$-algebra on $X$ in the proved bar-cobar regime. Then the universal twisting morphism $\tau\colon \overline{B}^{\mathrm{ch}}(\mathcal{A}) \to \mathcal{A}$ is a Maurer-Cartan element of* $\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$*, and the counit*

$$\epsilon\colon \Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$$

*is a quasi-isomorphism of curved dg chiral algebras.*

*Proof.* See [3, Thm. B]. The finite-type hypothesis reduces to finite-type vertex algebras, where the operadic bar-cobar machinery of Loday-Vallette [2] applies after the curved extension of Positselski [4]. □

## 3. The strong filtration axiom

**Definition 3.1** (Strong completion tower)**.** An augmented curved chiral $A_\infty$-algebra $\mathcal{A}$ on a curve $X$ is a *strong completion tower* if it carries a descending filtration

$$\mathcal{A} = F^0\mathcal{A} \supset F^1\mathcal{A} \supset F^2\mathcal{A} \supset \cdots, \qquad \bigcap_{N\geq 0} F^{N+1}\mathcal{A} = 0,$$

such that:

(1) $\mathcal{A}$ is separated and complete: $\mathcal{A} \cong \varprojlim_N \mathcal{A}_{\leq N}$, where $\mathcal{A}_{\leq N} := \mathcal{A}/F^{N+1}\mathcal{A}$;
(2) every quotient $\mathcal{A}_{\leq N}$ is of finite type and lies in the proved bar-cobar regime of Theorem 2.2;
(3) $\overline{\mathcal{A}} = F^1\mathcal{A}$: the augmentation ideal is the first filtration piece;
(4) all chiral $A_\infty$-operations are filtration-nondecreasing,

$$\mu_r(F^{i_1}\mathcal{A}, \ldots, F^{i_r}\mathcal{A}) \subset F^{i_1+\cdots+i_r}\mathcal{A}. \tag{3.1}$$

Axiom (3) ties the filtration to the augmentation. Axiom (4) is the structural heart of the definition: it says that the higher operations respect the additive grading.

*Remark* 3.2 (On the quotient maps)*.* Each projection $p_N\colon \mathcal{A}_{\leq N+1} \twoheadrightarrow \mathcal{A}_{\leq N}$ is automatically a strict morphism of curved chiral $A_\infty$-algebras, because the kernel $F^{N+1}\mathcal{A}/F^{N+2}\mathcal{A}$ is an $A_\infty$-ideal of $\mathcal{A}_{\leq N+1}$ by (3.1). The finite-stage bar constructions $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ therefore form an inverse system of curved dg chiral coalgebras with strict morphisms, and the completed bar construction is the literal inverse limit.

**Proposition 3.3** (Standard landscape examples)**.** *The following are strong completion towers with respect to the conformal-weight filtration* $F^N\mathcal{A} := \bigoplus_{h\geq N}\mathcal{A}_h$*:*

(a) *the affine Kac-Moody vertex algebra* $V_k(\mathfrak{g})$ *for any simple* $\mathfrak{g}$ *and any non-critical level* $k \neq -h^\vee$*;*
(b) *the Virasoro vertex algebra* $\mathrm{Vir}_c$ *for any* $c$*;*
(c) *the principal* $\mathcal{W}$*-algebra* $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ *for any simple* $\mathfrak{g}$ *and any non-critical* $k$*;*
(d) *the lattice vertex algebra* $V_\Lambda$ *for any positive-definite even lattice* $\Lambda$*.*

*Proof.* Each family admits a $\mathbb{Z}_{\geq 0}$-grading by $L_0$-eigenvalue with finite-dimensional graded pieces and vacuum at $h = 0$, so axioms (1)–(3) hold at once. For axiom (4), the bar differential extracts the coefficient of $(z-w)^{-1}$ in the OPE after desuspension, and this residue operation preserves the total weight on the augmentation ideal $\overline{\mathcal{A}} = F^1\mathcal{A}$ where all inputs have strictly positive weight. Iterating propagates the bound to $\mu_r(F^{i_1}, \dots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$.

Family-by-family: affine Kac-Moody has weight-1 generators $J^a$ with OPE poles of order $\leq 2$; Virasoro has a weight-2 generator $L$ with poles of order $\leq 4$; principal $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ has weight-$s$ generators for $s = 2, \dots, \mathrm{rk}(\mathfrak{g}) + 1$ with poles of order $\leq 2s$; and $V_\Lambda$ has weight-1 Cartan generators together with vertex operators $e^\alpha$ of weight $|\alpha|^2/2 \geq 1$. □

**Example 3.4** (Non-example: $\beta\gamma$-system with conformal grading)**.** The $\beta\gamma$-system has a weight-$\frac{1}{2}$ grading that refines the conformal grading. With respect to the conformal-weight filtration alone, the strong filtration axiom fails because the OPE $\beta(z)\gamma(w) \sim 1/(z-w)$ produces a weight-zero output from weight-$\frac{1}{2}$ inputs, violating additivity. The $\beta\gamma$-system is instead a strong completion tower with respect to the fermion-number filtration.

## 4. The degree cutoff lemma

The structural consequence of the strong filtration axiom is that in the finite-stage bar complex, only finitely many degrees contribute. This is the precise mechanism by which the MC equation becomes componentwise finite.

**Lemma 4.1** (Degree cutoff)**.** *Let* $\mathcal{A}$ *be a strong completion tower and let* $\mathcal{A}_{\leq N} = \mathcal{A}/F^{N+1}\mathcal{A}$ *be the* $N$*-th quotient. In the finite-stage convolution algebra* $\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N})$*, the Maurer-Cartan equation*

$$\partial(\tau_N) + \tau_N \star \tau_N = 0$$

*involves only degrees* $r \leq N$*: on* $\mathcal{A}_{\leq N}$ *the bar differential is the finite sum* $b_1 + b_2 + \cdots + b_N$ *of coderivations coming from* $\mu_1, \dots, \mu_N$*.*

*Proof.* Every bar input lies in $\overline{\mathcal{A}}_{\leq N}$, which by axiom (3) equals the image of $F^1\mathcal{A}$ in the quotient. The strong filtration axiom (3.1) gives

$$\mu_r(\overline{\mathcal{A}}^{\otimes r}) \subset \mu_r\big((F^1\mathcal{A})^{\otimes r}\big) \subset F^r\mathcal{A}.$$

Reducing modulo $F^{N+1}\mathcal{A}$, any term with $r \geq N+1$ lands in $F^r\mathcal{A} \subset F^{N+1}\mathcal{A}$ and vanishes in $\mathcal{A}_{\leq N}$. So the induced bar differential on $\mathcal{A}_{\leq N}$ is the finite sum $b_1 + b_2 + \cdots + b_N$, as claimed. □

*Remark* 4.2 (Degree cutoff as PBW degeneration). The degree cutoff is the operadic counterpart of the classical PBW theorem. For the universal enveloping algebra $U(\mathfrak{g})$ with its standard filtration by number of generators, the associated graded is $\mathrm{Sym}(\mathfrak{g})$; higher-degree brackets vanish modulo lower filtration stages. The strong filtration axiom extends this phenomenon to all chiral $A_\infty$-operations on a graded vertex algebra: the degree cutoff lemma is the statement that at each finite stage, only finitely many Poisson (or vertex) operations are active. The classical PBW theorem is the $r = 2$ case.

**Corollary 4.3** (Finite-stage MC assembly)**.** *Let $\mathcal{A}$ be a strong completion tower. The universal twisting morphism $\tau_N$ at stage $N$ is a Maurer-Cartan element of a* finite-dimensional *convolution $A_\infty$-algebra: the quotient $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ has finite-dimensional bar-length components, and the bar differential is the finite sum of Lemma 4.1.*

*Proof.* Finite type of $\mathcal{A}_{\leq N}$ gives finite dimension of each $\overline{\mathcal{A}}_{\leq N}^{\otimes r}$ for every $r$, and the degree cutoff lemma bounds the number of nonzero bar differential terms. Theorem 2.2 then gives the Maurer-Cartan property. □

## 5. Mittag-Leffler is automatic

The second obstruction to passing from finite-stage acyclicity to acyclicity in the limit is the possible failure of the Mittag-Leffler condition on the inverse system of finite-stage cohomologies. We show that the strong filtration axiom makes it automatic.

**Proposition 5.1** (Mittag-Leffler for strong towers)**.** *Let $\mathcal{A}$ be a strong completion tower. For every cohomological degree $m$, the inverse system*

$$\left\{H^m(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))\right\}_{N\geq 0}$$

*satisfies the Mittag-Leffler condition.*

*Proof.* The transition maps $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N+1}) \twoheadrightarrow \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ are surjections, because the finite-stage bar construction is functorial on surjections of finite-type curved chiral $A_\infty$-algebras and $p_N \colon \mathcal{A}_{\leq N+1} \twoheadrightarrow \mathcal{A}_{\leq N}$ is itself surjective.

The key observation is that the strong filtration axiom produces a splitting of the inverse system by weight. For each fixed conformal weight $w$, the weight-$w$ subcomplex of $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ involves only bar entries of total weight $w$, and such entries can only use generators of weight $\leq w$. Once $N \geq w$, all such generators already occur in $\mathcal{A}_{\leq N}$, so the transition map

$$\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N+1})_w \xrightarrow{\sim} \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})_w$$

is an isomorphism of subcomplexes. Passing to cohomology gives stable isomorphisms for $N \geq w$, hence the weight-$w$ cohomology slice stabilizes at $N = w$.

Summing over weights, the full cohomology $H^m(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))$ is a direct sum of finite-dimensional weight slices each of which stabilizes at a finite stage. In particular the transition maps are eventually isomorphisms on every fixed weight slice, so the inverse system is Mittag-Leffler: the derived limit $\varprojlim_N^1 H^m$ vanishes. □

**Lemma 5.2** (Complete filtered comparison)**.** *Let $\{f_N \colon C_N \to D_N\}$ be a morphism of inverse systems of complexes with surjective transition maps. If each $f_N$ is a quasi-isomorphism and the inverse systems $\{H^*(C_N)\}$ and $\{H^*(D_N)\}$ are Mittag-Leffler, then the induced map on limits $\widehat{f} \colon \widehat{C} \to \widehat{D}$ is a quasi-isomorphism.*

*Proof.* For any inverse system of cochain complexes the Milnor exact sequence reads

$$0 \to \varprojlim_N^1 H^{m-1}(C_N) \to H^m(\widehat{C}) \to \varprojlim_N H^m(C_N) \to 0.$$

Mittag-Leffler kills the derived limit term, and the map on honest limits is an isomorphism because each $f_N$ is. □

## 6. The main theorem

**Theorem 6.1** (Strong completion tower bar-cobar duality)**.** *Let $\mathcal{A}$ be a strong completion tower (Definition 3.1). Write $\tau_N \in \mathrm{MC}\big(\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N})\big)$ for the universal twisting morphism at stage $N$. Then:*

*(1)* Completed coalgebra. *The completed bar construction*

$$\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A}) := \varprojlim_N \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$$

*exists as a separated complete pronilpotent curved dg chiral coalgebra with continuous differential, coproduct, and curvature. At each stage, $\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})/F^{N+1} \cong \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$.*

*(2)* Completed MC element. *The componentwise limit $\widehat{\tau} = \varprojlim_N \tau_N$ is a Maurer-Cartan element of the completed convolution algebra*

$$\widehat{\mathrm{Conv}} := \varprojlim_N \mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N}),$$

*and the MC equation $\partial(\widehat{\tau}) + \widehat{\tau} \star \widehat{\tau} = 0$ converges in the inverse-limit topology, each component being a finite sum by the degree cutoff lemma.*

*(3)* Completed cobar. *The completed cobar object*

$$\widehat{\Omega}^{\mathrm{ch}}\big(\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})\big) := \varprojlim_N \Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))$$

*is a separated complete curved dg chiral algebra, equal to the twisted tensor product* $\mathcal{A} \otimes_{\widehat{\tau}} \widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})$.

(4) Counit is a quasi-isomorphism. *The completed counit*

$$\widehat{\epsilon}\colon \widehat{\Omega}^{\mathrm{ch}}\big(\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})\big) \xrightarrow{\sim} \mathcal{A}$$

*is a quasi-isomorphism of curved dg chiral algebras.*

(5) Dual quasi-isomorphism. *If* $\mathcal{C} = \varprojlim_N \mathcal{C}_{\leq N}$ *is a separated complete pronilpotent curved dg chiral coalgebra whose finite quotients lie in the theorematic regime, then the completed unit*

$$\widehat{\eta}\colon \mathcal{C} \xrightarrow{\sim} \widehat{\overline{B}}^{\mathrm{ch}}\big(\widehat{\Omega}^{\mathrm{ch}}(\mathcal{C})\big)$$

*is a quasi-isomorphism.*

(6) Quasi-inverse equivalences. *Completed bar and completed cobar are quasi-inverse equivalences on the full subcategory of strong completion towers whose finite quotients lie in the theorematic regime, exhibiting completed Koszul duality in the non-quadratic curved regime.*

*Proof. Step 1: the completed coalgebra.* Each quotient map $p_N\colon \mathcal{A}_{\leq N+1} \twoheadrightarrow \mathcal{A}_{\leq N}$ is strict by axiom (4), inducing a strict morphism $q_N\colon \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N+1}) \to \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ compatible with differential, coproduct, coaugmentation, and curvature. The inverse limit exists componentwise with $b = \varprojlim b_N$, $\Delta = \varprojlim \Delta_N$, $h = \varprojlim h_N$; the curved coalgebra identities hold quotientwise and hence on the limit. Continuity is formal for the inverse-limit topology. Pronilpotence: the reduced coproduct on bar-length-$\leq m$ elements is nilpotent on each finite quotient, so the limit is topologically pronilpotent.

*Step 2: MC assembly.* Define $\widehat{\tau} := \varprojlim_N \tau_N$. By Lemma 4.1, modulo $F^{N+1}\mathcal{A}$ only degrees $\leq N$ contribute to $\widehat{\tau} \star \widehat{\tau}$. The MC equation holds modulo $F^{N+1}$ (where it reduces to the finite-stage equation for $\tau_N$), hence on the completed coalgebra. This is claim (2).

*Step 3: completed cobar.* Each $\Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))$ is a curved dg chiral algebra; compatible transition maps give a complete curved dg chiral algebra in the limit, with continuity automatic. This is claim (3).

*Step 4: counit is a quasi-isomorphism.* The quotient of $\widehat{\epsilon}$ modulo $F^{N+1}$ is the finite-stage counit $\epsilon_N$, which is a quasi-isomorphism by Theorem 2.2. By Proposition 5.1, the inverse system $\{H^m(\mathcal{A}_{\leq N})\}$ is Mittag-Leffler. Applying Lemma 5.2 to the morphism $\{\epsilon_N\}$ of inverse systems yields claim (4).

*Step 5: dual quasi-isomorphism.* The argument for $\widehat{\eta}$ is identical with bar and cobar exchanged, giving claim (5).

*Step 6: quasi-inverse equivalences.* Claims (4) and (5) are the two triangle identities up to quasi-isomorphism, so completed bar and completed cobar form an adjoint equivalence on the homotopy categories restricted to strong completion towers. This is claim (6). □

## 7. Positive versus resonant: the MC4 dichotomy

Theorem 6.1 upgrades the MC4 completion closure from a conjecture to a theorem with sharp hypotheses. The conditions split the completion landscape into two regimes.

### 7.1. Positive towers (MC4$^+$).

A strong completion tower is *positive* if the filtration comes from a genuine $\mathbb{Z}_{\geq 1}$-grading preserved by all $A_\infty$-operations, so axiom (4) holds with equality.

**Corollary 7.1** (MC4$^+$ is closed)**.** *For any positive strong completion tower $\mathcal{A}$, completed bar-cobar duality holds unconditionally: the counit $\widehat{\epsilon}$ and the unit $\widehat{\eta}$ are quasi-isomorphisms.*

*Proof.* Positive towers are strong completion towers, so Theorem 6.1 applies. □

This closes the MC4$^+$ problem for all positive towers, including $\mathcal{W}_{1+\infty}$, affine Yangians with positive filtration, and positive RTT towers.

### 7.2. Resonant towers (MC4$^0$).

A strong completion tower is *resonant* if there exists a finite-dimensional weight-zero piece $R_{\mathcal{A}} \subset \mathcal{A}$ on which higher operations can preserve filtration degree. The *resonance rank* is $\rho(\mathcal{A}) := \dim H^*(R_{\mathcal{A}}, m_1^{R_{\mathcal{A}}})$. For $\rho = 0$ the tower is positive and Corollary 7.1 applies. For $0 < \rho < \infty$ (the *finite resonance regime*) the strong filtration axiom still holds on the ambient tower and Theorem 6.1 produces a completed bar-cobar quasi-isomorphism with extra cohomology concentrated in the resonance sector. For $\rho = \infty$ the tower is wild and lies outside the present framework.

**Theorem 7.2** ($W_N$ rigidity)**.** *Let $\mathcal{A} = W_N$ be the principal finite-type $\mathcal{W}$-algebra with generators of conformal weights $2, 3, \ldots, N$. The standard conformal-weight filtration makes $W_N$ into a strong completion tower in the finite resonance regime. For every cohomological degree $m$ and every conformal weight $w \geq 0$, the transition map*

$$H^m(\overline{B}^{\mathrm{ch}}(W_{N+1}))_w \xrightarrow{\sim} H^m(\overline{B}^{\mathrm{ch}}(W_N))_w$$

*is an isomorphism for all $N \geq w$. Consequently the tower*

$$\mathcal{W}_\infty := \varprojlim_N W_N$$

*admits a completed bar-cobar duality: the counit $\widehat{\epsilon}_{\mathcal{W}_\infty}$ is a quasi-isomorphism.*

*Proof.* $W_N$ satisfies the hypotheses of Proposition 3.3(c) at each finite stage. The transition map $W_{N+1} \to W_N$ forgets the generator of weight $N+1$, which is a strict morphism of curved chiral $A_\infty$-algebras. A bar chain of total conformal weight $w$ can involve only generators of weight $\leq w$; once $N \geq w$, all such generators already occur in $W_N$, so the weight-$w$ subcomplex is identical in every later stage.

Proposition 5.1 then yields the Mittag-Leffler property on each weight slice, and Theorem 6.1 gives completed bar-cobar duality for the limit $\mathcal{W}_\infty$. The finite resonance rank is computed from the scalar part of the $W$-OPEs at the weight-zero resonance sector and is finite at every stage. □

*Remark* 7.3 (Twenty-one conjectures resolved). Theorem 7.2 resolves twenty-one standing conjectures in the completion frontier for principal $\mathcal{W}$-algebras, catalogued in the monograph as the $W_N$ rigidity conjectures for $N = 2, \ldots, 8$ together with the asymptotic stabilization, pro-nilpotence, surjectivity, and uniqueness conjectures. Each of these conjectures asked for a specific completion property; Theorem 7.2 produces them all as consequences of the strong filtration axiom.

## 8. Examples

8.1. **$\beta\gamma$ and affine Kac-Moody.** Under the fermion-number filtration the $\beta\gamma$-system is a positive strong completion tower ($\rho = 0$): the single-pole OPE $\beta(z)\gamma(w) \sim 1/(z-w)$ gives a Heisenberg-type contribution at bar degree 2 and no higher-degree operations. Under the conformal grading, the strong filtration axiom fails; the fermion-number refinement is essential.

For $V_k(\mathfrak{g})$ at non-critical level the conformal-weight filtration makes $V_k(\mathfrak{g})$ into a positive strong completion tower: the $J^a \cdot J^b$ OPE has poles of order at most 2, so the binary bar operation has weight shift $\leq 1$ and Corollary 7.1 applies directly. At the critical level $k = -h^\vee$ the weight-zero resonance becomes infinite and the framework does not apply.

8.2. **Virasoro, $\mathcal{W}_3$, and the principal series.** Virasoro at central charge $c$ has a single weight-2 generator $L$ with a quartic pole in $T \cdot T$:

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

The conformal-weight filtration makes $\mathrm{Vir}_c$ into a strong completion tower with weight-zero resonance given by the central term $c/2$. The resonance rank is $\rho = 1$, so $\mathrm{Vir}_c$ lies in the finite resonance regime and Theorem 6.1 gives a completed bar-cobar quasi-isomorphism.

The $\mathcal{W}_3$ algebra has generators $T$ (weight 2) and $W$ (weight 3); the $W \cdot W$ OPE has a sixth-order pole, hence higher-degree bar operations than Virasoro. The strong filtration axiom still holds; the resonance rank is $\rho = 2$. The same argument handles the entire principal series $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$, and Theorem 7.2 gives rigidity at the level of the tower $\mathcal{W}_\infty = \varprojlim_N W_N$.

8.3. **What is excluded.** The strong filtration axiom fails for the $\beta\gamma$-system under the raw conformal grading, for affine $\widehat{\mathfrak{g}}$ at critical level, for algebras with non-additive spectral parameters, and for the open sector of Swiss-cheese algebras where operations act across two colors. A different filtration can sometimes restore axiom (4), but the raw conformal grading alone is insufficient.

8.4. **Summary table.**

| Algebra | Filtration | Strong? | $\rho$ | Regime |
|---|---|---|---|---|
| $V_k(\mathfrak{g})$, $k \neq -h^\vee$ | conformal | yes | 0 | $\mathrm{MC4}^+$ |
| $V_\Lambda$ | conformal | yes | 0 | $\mathrm{MC4}^+$ |
| $\beta\gamma$ | fermion number | yes | 0 | $\mathrm{MC4}^+$ |
| $\beta\gamma$ | conformal | no | n/a | n/a |
| $\mathrm{Vir}_c$ | conformal | yes | 1 | $\mathrm{MC4}^0$ |
| $\mathcal{W}_3$ | conformal | yes | 2 | $\mathrm{MC4}^0$ |
| $W_N$ | conformal | yes | $< \infty$ | $\mathrm{MC4}^0$ |
| $V_{-h^\vee}(\mathfrak{g})$ | conformal | yes | $\infty$ | wild |

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